Null and Alternative Hypotheses

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Introduction

Every hypothesis test begins with two hypotheses: a **null** that captures the status quo and an **alternative** that captures what we would conclude if the null is rejected. Stating both clearly, before the data arrive, is the crucial first step – and a remarkably common source of error when skipped.

Prerequisites

Random variables and population parameters.

Theory

The **null hypothesis** H_0 is typically a statement of no effect, no difference, or a specific parameter value. The **alternative** H_1 (or H_A) is what we would conclude if we reject H_0.

Two-sided alternatives are the default when the sign of the effect is unknown or of interest in both directions:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0.$$

One-sided alternatives specify a direction, increasing power for that direction at the cost of giving up inference in the opposite:

$$H_0 : \mu \leq \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0.$$

Use a one-sided test only when:

- A priori, only one direction is of scientific interest.
- A result in the other direction would be treated identically to a null result.
- The directionality is pre-specified in the protocol, not chosen after seeing the data.

Simple vs. composite: $H_0 : \mu = 0$ is simple (a single value); $H_0 : \mu \leq 0$ is composite (a set). Test statistics and rejection rules differ slightly.

Equivalence/non-inferiority testing flips the roles: the null is that the effect is at least as large as some margin, and rejection establishes equivalence.

Assumptions

Hypotheses must be specified before examining the data. Post-hoc redefinition invalidates the sampling distribution of the p-value.

R Implementation

```r
set.seed(2026)

# Two-sided test
x <- rnorm(40, mean = 2, sd = 5)
t.test(x, mu = 0, alternative = "two.sided")

# One-sided test (H1: mu > 0)
t.test(x, mu = 0, alternative = "greater")

# One-sided test (H1: mu < 0)
t.test(x, mu = 0, alternative = "less")

# TOST (equivalence: |mu| < 1)
library(TOSTER)
tsum_TOST(m1 = mean(x), sd1 = sd(x), n1 = length(x),
          low_eqbound = -1, high_eqbound = 1)
```

Output & Results

The two-sided p will be roughly twice the one-sided p when the point estimate is consistent with H_1's direction. For equivalence testing, rejection of both one-sided components establishes equivalence.

Interpretation

Pre-register or pre-specify the hypotheses in the protocol; do not switch from two-sided to one-sided after seeing the sign of the effect. Misuse of one-sided tests is a common source of reviewer objections.

Practical Tips

- Default to two-sided tests unless a strong, pre-specified rationale for one-sidedness exists.
- Never compute both one-sided and two-sided p-values and report whichever is smaller.
- In equivalence trials, the "null" is dissimilarity and "alternative" is equivalence – reversed from classical tests.
- Composite null hypotheses are handled by evaluating the test statistic at the boundary.
- Frame hypotheses in terms of the parameter of scientific interest, not the test statistic.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests

- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests